MILITARY SPECIFICATION

PENETRATORS, BULKHEAD, CONNECTORIZED, FIBER OPTIC, (FOR INBOARD USE ON NAVY SHIPS AND SUBMARINES) GENERAL SPECIFICATION FOR

This specification is approved for use by Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and characteristics for fiber optic connectorized bulkhead penetrators intended for the penetration of inboard pressure-proof bulkheads on navy ships and submarines with fiber optic cables utilizing one or more single-fiber transmission elements. Penetrators are compatible with cable specified in DOD-C-85045(SH).

1.2 Classification. Fiber optic connectorized bulkhead penetrators covered by this specification shall be the following types as specified (see 6.2).

Type I - Receptacle mounting.

Type II - Plug.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2) the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-465	-	Copper - Aluminum Alloys (Aluminum Bronze) (Copper Alloy Numbers 606, 614, 630, 623M, and 642); Rod, Flat Products with Finished Edges (Flat Wire, Strip, and Bar), Shapes and Forgings.
QQ-P-35	-	Passivation Treatments for Corrosion-Resisting Steel.
QQ-S-763	-	Steel Bars, Wire, Shapes and Forgings, Corrosion-Resisting.

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MIL-S-901	-	Shock Tests, H. I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Sea Systems Command (SEA 55Z3), Department of the Navy, Washington, DC 20362, by using the self-addressed Standardization Documents Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-W-22759/43	-	Wire, Electrical, Fluoropolymer Insulated, Crosslinked Modified ETFE, Normal Weight Silver Coated Copper, 200°C, 600 Volt.
MIL-P-24627/1	-	Penetrator, Bulkhead, Connectorized, Fiber Optic 32 Channel (For Inboard use on Navy Ships and Submarines).
MIL-P-25732	-	Packing, Preformed, Petroleum Hydraulic, Fluid Resistant, 275°F.
MIL-C-55330	-	Connectors, Electrical and Fiber Optic, Packaging of.

STANDARDS

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MIL-STD-104	-	Limits for Electrical Insulation Color.
MIL-STD-454	-	Standard General Requirements for Electronic Parts.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MIL-STD-1344	-	Test Methods for Electrical Connectors.
DOD-STD-1678	-	Fiber Optics Test Methods and Instrumentation.
MIL-STD-45662	-	Calibration Systems Requirements.
MS20995	-	Wire, Safety or Lock.
MS28775	-	Packing Preformed, Hydraulic 275°F O-Ring.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified, (see 6.2), samples shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Materials. The bulkhead penetrators shall be constructed of materials as specified herein (see 3.1). All materials used shall be ozone resistant (see 3.7.7), non-nutrient to fungus (see requirement 4 of MIL-STD-454), and manufactured to good workmanship quality (see 3.9). Bulkhead penetrator materials shall provide the degree of immunity to nuclear radiation exposure as specified (see 3.1).

3.3.1 Toxic products and formulations. Materials used in bulkhead penetrator construction shall not give off toxic or explosive fumes when exposed to flame. Elemental mercury or asbestos shall not be used. The material used shall have no adverse effect on the health of personnel when used for its intended purpose.

3.3.2 Nonmagnetic materials. Unless otherwise specified herein, all component parts shall be of a nonferrous material or a material generally considered to be nonmagnetic.

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3.3.3 Component materials.

3.3.3 1 Receptacle bodies. Receptacle body material shall be of corrosion-resistant steel, class 316 of QQ-S-763, and shall be passivated as specified in QQ-P-35.

3.3.3.2 Plug sleeves. Sleeves shall be fabricated from corrosion-resistant steel, class 316 of QQ-S-763, and shall be passivated as specified in QQ-P-35.

3.3.3.3 Coupling rings. Coupling rings shall be fabricated from nickel aluminum bronze in accordance with copper alloy No. 630 of QQ-C-465.

3.3.3.4 Termini. Optical termini used for retaining and alining the optical fiber, shall be made of noncorrosive material of sufficient hardness to meet the specified durability requirement (see 3.7.9) for repeated connector matings. Optical termini shall provide long term, in service, optical fiber alinement for sustained optical performance across the connector junction. The material used shall in no manner interfere with or degrade the fiber optical termination process, terminus cleaning operation or optical junction transmission. Plating which might provide a source of contamination to the optical junction under conditions of repeated matings shall not be employed on these termini. When epoxy or a comparable cement is employed in the fiber to terminus securing or polishing processes, there shall be no incompatibilities between the materials employed such that degradation of these materials would result from in-service use or from the temperature life requirements of 3.7.2.

3.3.3.5 Inserts. Insert materials will require adequate hardness and wear resistance (see 3.7.9) to retain optical terminus alinement over repeated mating operations. Inserts may be of dielectric or conductive material as required.

3.3.3.6 Protective covers. Penetrator plugs and receptacles shall be provided with protective covers to prevent exposure and damage to the connectors when the mating connectors are not installed. Protective covers shall be molded or machined from Styrene - ABS, or an alternate plastic or nonferrous material.

3.3.3.7 Lock wire. Lock wire, if specified (see 3.1), shall be of nickel-copper (Monel) conforming to MS20995.

3.3.3.8 Seals. O-ring seals shall be molded of Buna N rubber in accordance with MIL-P-25732.

3.3.4 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Design and construction. The bulkhead penetrators shall be designed and constructed as follows in this document.

3.4.1 Detailed requirements. Detailed design requirements for the penetrators specified herein shall include means to accomplish the following:

a. Permit establishment and preservation of the bulkhead pressure-proof (see 6.4.7) integrity.

b. Permit tensile strength continuity between cable and penetrator without application of tensile load to the termini.

c. Permit environmental sealing of the optical junction(s) to the entry of all external agents.

d. Permit establishment and preservation of the optical alinement between mating connectors as specified in terms of their optical insertion loss requirement.

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3.4.2 Termini. The optical terminus dimensions shall be as specified (see 3.1). The termini shall provide a scoop-proof capability between mating connectors. The design shall contain and locate the optical fiber(s) within the specified interface geometry. Termini used for this purpose shall conform to 3.3.3.4. The mating ends of these termini shall be shaped to prevent gouging between a terminus and its insert channel during engagement. For those connectors employing removable termini, the termini shall prevent damage to the terminus retention device or sealing member(s) during insertion or removal of a terminus. No terminus damage or optical junction contamination shall result from normal twisting or forcing actions developed during connector mating and unmating operations; or by insertion of termini into or removal of termini from the connector with the tool specified for insertion or removal. Terminus cleaning capability shall conform to the visual examination of 4.6.1.2 and shall meet the insertion loss requirements of 3.6.1 when tested in accordance with 4.6.3.20.

3.4.3 Inserts. The inserts shall be secured to prevent rotation within the shell. The insert design and construction shall preserve the optical termini environmental sealing and alinement and must be capable of ensuring sustained optical performance as specified (see 3.1).

3.4.4 Shells. The shells shall retain their inserts as specified in 3.4.3. The shells shall be designed to provide cable support as specified (see 3.1).

3.4.5 Receptacles. The receptacle design and dimensions shall be as specified (see 3.1). The receptacle shall be keyed to accept mating shells. A positive means shall be provided to prevent rotation of the receptacle with respect to the bulkhead.

3.4.6 Engagement of connectors. Counterpart connectors and accessories shall be capable of being fully engaged and disengaged without excessive binding or galling. A positive locking mechanism such as lockwire shall be provided as specified (see 3.1).

3.4.7 Cable sealing. Cables terminating into a penetrator plug or receptacle shall be sealed in such a manner as to withstand sea water pressures required by 4.6.3.8.

3.4.8 O-Ring seals. O-ring seals shall conform dimensionally to MS28775.

3.4.9 Interchangeability. All penetrator plugs and receptacles having the same military part number shall be physically and functionally interchangeable without need for modification of such items or of the mating equipment.

3.4.10 Dissimilar metals. Intimate contact between dissimilar metals shall be avoided, where possible. Should contacting dissimilar metals be employed in a penetrator, suitable protection against electrolytic corrosion shall be provided as specified in MIL-STD-889.

3.5 Transportability. Packing of bulkhead penetrator plugs, receptacles, and accessories shall be as specified in section 5 herein to permit universal transportability of these products by all carriers without need for special support provisions.

3.6 Optical performance. Optical tests shall be used to monitor effects of the requirements as specified in 3.7 and 3.8 in accordance with 4.4.2 and 4.5. Three channels or 20 percent of the channels, whichever is larger, should be used when dealing with the number of channels.

3.6.1 Insertion loss. When measured in accordance with 4.6.2.2, the maximum per channel insertion loss shall be 3.0 decibels at the wavelength as specified (see 3.1).

3.6.2 Discontinuity. For those tests specifying discontinuity assessment during test exposure, and when measured in accordance with 4.6.2.3, no reduction in signal strength of 2 decibels or more for a duration of 20 nanoseconds or more shall occur.

3.6.3 Analog modulation. When tested in accordance with 4.6.2.4, the peak-to-peak analog modulation, bandpass limited to between 4 hertz and 40 kilohertz, shall be not more than 1 percent of the steady state signal level unless otherwise specified (see 3.1).

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3.6.4 Pulse distortion. When tested in accordance with 4.6.2.5, the rise and fall times between the 10 and 90 percent pulse optical power amplitude levels shall be no more than 0.5 nanosecond longer than the rise and fall times of the incident pulses. The duration of each of the transmitted pulses shall be no more than 0.5 nanoseconds greater than the duration of each of the incident pulses.

3.6.5 Crosstalk. When tested in accordance with 4.6.2.6, and unless otherwise specified (see 3.1), the output signal optical power level or sum of levels, for devices with three or more channels, of the passive channel or channels shall be below the output signal level of the active channel by at least 60 decibels.

3.6.6 Ambient optical pickup. When tested in accordance with 4.6.2.7, the optial power of the light from the optical port(s) (after accounting for cable and optical junction losses between the device and the detector) shall be less than -70 dBm.

3.7 Environmental requirements.

3.7.1 Thermal shock. When tested in accordance with 4.6.3.1, the penetrators subjected to the high and low temperature extremes shall reveal no evidence of connector part dimensional change, no apparent loss of sealing capability, no surface or identification marking impairment, no coupling-thread binding or other evidence of mating or unmating incapability and no other damage detrimental to the operation of the connector. The optical performance requirements of table I shall be met.

3.7.2 Temperature life. When tested in accordance with 4.6.3.2, the penetrators subjected to these specified accelerated aging exposures shall not exhibit visual evidence of dimensional change, opening or leaking of seals, cracking or crazing of components or finishes, identification marking impairment, or other effects detrimental to their operation. The optical performance requirements of table I shall be met.

3.7.3 Humidity. When tested to high humidity and cyclic temperature exposures in accordance with 4.6.3.3 the penetrator parts shall not swell, impair identification marking or otherwise degrade such that the mechanical design requirements of 3.4 cannot be met. The optical requirements of table I shall be met.

3.7.4 Salt spray (corrosion). When tested in accordance with 4.6.3.4, no visible evidence of salt penetration into the penetrator sealed area shall be observed. No corrosive effects shall be evident on the external connector parts. The optical performance requirements of table I shall be met.

3.7.5. Fluid immersion. When tested in accordance with 4.6.3.5, visual examination of the penetrator shall reveal no swelling or softening of material, no loss of sealing capability or identification marking and no discoloration or other effects detrimental to the intended use of these penetrators. The optical performance requirements of table I shall be met.

3.7.6 Flammability. When tested in accordance with 4.6.3.6, penetrator burning after removal of the applied flame and after-glow shall both extinguish within the time specified in MIL-STD-1344 method 1012 test condition A of table I and test condition A of table II. There shall be no dripping which would cause flammable material to ignite. There shall be no violent burning or explosive type fire (see 3.3.1).

3.7.7 Ozone exposure. When tested in accordance with 4.6.3.7, the seals shall show no evidence of excessive swelling which may degrade environmental isolation. The optical performance requirements of table I shall be met.

3.7.8. Hydrostatic pressure. When tested in accordance with 4.6.3.8, the penetrator receptacles shall reveal no permanent dimensional changes; no cracking or crazing of insulation material; no permanent damage of cable sealing compounds; no leakage of receptacle-to-bulkhead seals, plug-to-receptacle seals, and cable-to-plug seals; no displacement or rupture of connector shells, inserts, termini, fibers, or sealing compounds or any other damage which may degrade the performance of the penetrator receptacle.

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3.7.9 Mating durability. When tested in accordance with 4.6.3.9, the mating penetrator plugs and receptacles shall show no evidence of excessive wear on engaging hardware, no uneven wear or galling on guide hardware, no metal chips or filings in the optical contact area, no damaged inserts, nor any other mechanical defects detrimental to connector operation. The optical performance requirements of table I shall be met.

3.7.10 Terminus retention. When tested in accordance with 4.6.3.10, and subjected to axial compressive loads up to 100 newtons, the termini shall be retained in their inserts.

3.7.11 Cable seal flexing. When tested in accordance with 4.6.3.11, the penetrator plug or receptacle strain relief and pressure-proof molded cable seals shall prevent loss of environmental sealings or other damage which may impair the connector operation. The optical performance requirements of table I shall be met.

3.7.12 Twist. When tested in accordance with 4.6.3.12, the plug or receptacle seals shall not be rendered inoperable for their intended purpose nor shall any other connector damage occur which would degrade the operation of the penetrator. The optical performance requirements of table I shall be met.

3.7.13 External bending moment. When tested in accordance with 4.6.3.13, the connectors shall exhibit no evidence of damage to seals or plug shells. The optical performance requirements of table I shall be met.

3.7.14 Cable pullout. When tested in accordance with 4.6.3.14, the penetrator shall show no signs of cable jacket damage, fiber damage, cable to penetrator seal damage, cable to pressure-proof molded cable seal or other damage which would degrade the performance of the penetrator. The optical performance requirements of table I shall be met.

3.7.15 Impact (drop). When tested in accordance with 4.6.3.15, the penetrator plugs shall not be visibly damaged or otherwise rendered unfit for operational use. The optical performance requirements of table I shall be met.

3.7.16 Mechanical shock. When tested in accordance with 4.6.3.16, the penetrators shall not be damaged and there shall be no loosening of parts. The optical performance requirements of table I shall be met.

3.7.17 Vibration. When tested in accordance with 4.6.3.17, the penetrators shall exhibit no visual evidence of loosening of parts, relative motion between penetrator and cable parts or other damage which can produce physical distortion and result in fatigue of the mechanical parts. The optical performance requirements of table I shall be met.

3.7.18 Terminus insertion and removal forces. When tested in accordance with 4.6.3.18, the terminus insertion forces and the forces required to remove unlocked termini shall not exceed 100 newtons.

3.7.19 Water pressure. When tested in accordance with 4.6.3.19, visual inspection of the penetrator shall reveal no penetration of indicator dye into the sealed region of the penetrator.

3.8 Physical properties.

3.8.1 Size. When examined in accordance with 4.6.1, the dimensions of the penetrator parts shall be in accordance with the values as specified (see 3.1).

3.8.2 Mass. When tested in accordance with 4.6.4.1, the mass of the penetrator parts shall not exceed the value as specified (see 3.1).

3.8.3 Color. The external color of the penetrator(s) shall be as specified in the acquisition document. When examined in accordance with 4.6.4.2 colors shall conform to MIL-STD-104, class 1.

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3.8.4 Identification marking. Except as otherwise specified in the acquisition document as specified (see 3.1) when examined in accordance with 4.6.4.3 the assembled penetrator and accessories shall be identified by a legible and permanent marking applied to the protective cover, shell, or coupling mechanism in accordance with MIL-STD-1285 except marking characters shall be a minimum of 2 millimeters in height.

3.8.5 Marking durability. The marking characters on any face of the penetrator part or enclosure shall remain legible after completion of the test specified in 4.6.1.

3.9 Workmanship. The penetrators shall be dimensionally uniform and free of manufacturing flaws that would degrade performance after installation, would inhibit proper connection to interfacing elements, and would otherwise yield an inferior product. The following shall be a minimum level of visual examination to be performed and is not intended to restrict other pertinent workmanship examinations deemed necessary by the contractor.

a. Penetrators and accessories shall conform to the dimensional and interchangeability requirements as specified herein.

b. Loose termini, poor molding fabrication, loose materials, defective bonding, damaged or improperly assembled termini and physical defects in the hermetic seals shall not be permitted.

c. Peeling or chipping of plating or finish, galling of mated parts, nicks or burrs of metal parts, and post molding warpage shall not be permitted.

d. Connector identification marking shall be legible and as specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Provision for test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection(s) shall be the responsibility of the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662 and as specified herein.

4.1.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the following test conditions:

a. Temperature: 15°C to 35°C.

b. Atmospheric pressure: 550 to 800 millimeters of mercury.

c. Relative humidity: 20 to 80 percent.

4.2 Classification of inspection. The inspection and testing of penetrators covered by this specification are classified as follows:

1. Materials inspection (see 4.3).

2. First article inspection (see 4.4).

3. Quality conformance inspection (see 4.5).

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4.3 Materials inspection. Materials inspection shall consist of certification, supported by verifying data, that materials specified (see 3.1 and 3.3) were used in fabricating the delivered penetrator.

4.3.1 Toxicological data and formulations. The contractor shall furnish to the contracting activity the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

4.4 First article inspection. First article inspection shall be performed on sample units produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts. All penetrator assemblies or parts submitted for first article acceptance testing shall be examined as specified herein.

4.4.1 Sample size. Six receptacles of each shell size as specified (see 3.1) shall be provided; two for the unmated and four for the mated testing. Four plug assemblies of each shell size shall be provided with one plug assembly of each size cabled and molded with appropriate cable. The samples shall be subjected to the tests of table I.

4.4.2 Inspection routine. Penetrator samples shall be tested in the sequence as specified in table I. Optical performance assessments shall be made as specified before, during, or after each group test exposure. All sample units shall be subjected to the inspection of group I. The quantity of test penetrator samples shall be as identified in table I.

4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspection. Quality conformance inspection shall be performed on 100 percent of production parts. Two samples shall be tested and no failures shall be allowed.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall follow the test sequence in table II.

4.5.2 Noncompliance. If a unit fails to pass the inspection, the manufacturer shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc.,and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the procuring activity has been taken. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.3 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-55330.

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TABLE I. First article inspection.

Examination or test	Optical tests: Insertion loss 3.6.1	Discontinuity 3.6.2	Analog modulation 3.6.3	Pulse distortion 3.6.4	Crosstalk 3.6.5	Ambient optical pickup 3.6.6	Requirement paragraph	Method paragraph
Group I (all sample units)								
Physical conformance								
Size							3.8.1	4.6.1
Mass							3.8.2	4.6.4.1
Color							3.8.3	4.6.4.2
Identification marking							3.8.5	4.6.4.3
Workmanship							3.9	4.6.1
Optical conformance								
Insertion loss							3.6.1	4.6.2.2
Pulse distortion							3.6.4	4.6.2.5
Crosstalk							3.6.5	4.6.2.6
Ambient optical pickup							3.6.6	4.6.2.7
Group II (2 sample units)								
Temperature life	A 1/						3.7.2	4.6.3.2
Humidity	A						3.7.3	4.6.3.3
Fluid immersion	A						3.7.5	4.6.3.5
Terminus insertion and removal forces							3.7.18	4.6.3.18
Terminus retention force							3.7.10	4.6.3.10
Color							3.8.3	4.6.4.2

See footnotes at end of table.

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TABLE I. First article inspection - Continued.

Examination or test	Optical tests						Requirement paragraph	Method paragraph
	Insertion loss 3.6.1	Discontinuity 3.6.2	Analog modulation 3.6.3	Pulse distortion 3.6.4	Crosstalk 3.6.5	Ambient optical pickup 3.6.6		
Identification marking							3.8.4	4.6.4.3
Group III (2 sample units)								
Thermal shock	A						3.7.1	4.6.3.1
Vibration	A	D 2/	D				3.7.17	4.6.3.17
Mechanical shock	A						3.7.16	4.6.3.16
Twist	A						3.7.12	4.6.3.12
Ozone exposure	A						3.7.7	4.6.3.7
Cable seal flexing	A						3.7.11	4.6.3.11
External bending moments	A	D	D				3.7.13	4.6.3.13
Cable pullout	DA						3.7.14	4.6.3.14
Water pressure	DA			A	A	A	3.7.19	4.6.3.19
Hydrostatic pressure	A						3.7.8	4.6.3.8
Group IV (2 sample units)								
Mating durability	A						3.7.9	4.6.3.9
Salt spray	A						3.7.4	4.6.3.4
Terminus cleaning	A						3.4.2	4.6.3.20
Impact	A			A	A	A	3.7.15	4.6.3.15
Flammability							3.7.6	4.6.3.6

1/ A - Optical tests performed after the environmental test.
2/ D - Optical tests performed during the environmental test.

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TABLE II. Quality conformance inspection.

Examination or test	Requirement paragraph	Method paragraph
Examination of product	3.8	4.6.1
Hydrostatic pressure	3.7.8	4.6.3.8
Insertion loss	3.6.1	4.6.2.2

4.6 Methods of inspection.

4.6.1 Visual and mechanical examinations. The penetrators, accessories and piece parts shall be examined to verify that the design, construction, physical characteristics and dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.8, and 3.9). Attention shall be given to those assemblies requiring sealing mechanisms as to the condition of the supplied seal. Missing, twisted, buckled, kinked or otherwise damaged seals shall be cause for rejection. A number of mechanical property tests specified herein require the visual inspection of the test-exposed, interior and exterior regions of the penetrator assembly to assess material damage or contamination conditions.

4.6.1.1 Plug accessories. Plug accessories shall be manually mated and unmated 5 times to their counterpart uncabled plugs. No evidence of excessive thread binding, seal pinching or any contamination build-up shall be observed after this procedure is completed.

4.6.1.2 Terminus cleaning. Cabled penetrators of each mateable type used shall be tested to examine the mechanism or procedure whereby the optical terminus faces are made accessible for cleaning. Ready access to these faces shall be noted, and no means shall be seen for containment of contaminating materials in the optical junction areas. These same penetrators shall be utilized to perform the terminus cleaning test of 4.6.3.20.

4.6.2 Optical test methods. For those tests where optical assessment of a penetrator performance property is specified, the penetrator test assemblies shall follow the test sequence in table I, selecting the same connector termini for all succeeding tests. These penetrator test assemblies shall be prepared with two meter lengths of the specified cable types with polished terminals on the ends opposite the penetrator. Suitable fiber end preparation shall be employed using the procedures outlined in method 8040 of DOD-STD-1678. Cladding mode stripping devices shall be inserted in the test measurement circuit between the source and the connector (penetrator input) and also between the connector (penetrator input) and the detector.

4.6.2.1 Number of channels. The number of channels shall be as specified (see 3.1) and determined by light transmission through the penetrator assembly and direct visual count of the optically continuous elements.

4.6.2.2 Insertion loss. A suitable optical power source such as an LED or ILD shall be used to launch optical power into an optical circuit containing the device under test. Continuous, tuned or synchronous power measurement techniques as described in Method 6010 of DOD-STD-1678 may be used. Cladding mode stripping devices shall be used to remove cladding modes from both the input and output signals of the penetrator channel(s) undertest. The insertion loss in decibels is $10 \log_{10} P_i/P_o$ where P_i is the total optical signal power into the device under test, and P_o is the total optical signal power out of the device under test. To insure that the optical loss calculated is due solely to the insertion of the device under test, the data obtained from the transmission path containing the device under test shall be corrected to yield the true power directed into and out of the device. This may be accomplished by either providing optical circuit stability or by using data from a reference optical transmission path to correct the data obtained from the transmission path containing the penetrator.

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4.6.2.3 Discontinuity. The optical ports of the device under test shall be connected through short optical test cables to an optical signal source and detector. The source shall produce a constant level, static signal easily detected by the detector. The output of the detector is monitored for discontinuities while the device under test is submitted to test exposures. The monitoring equipment should include suitable transient capture capability (such as provided by an oscilloscope with an adjustable sweep trigger and oscilloscope camera).

4.6.2.4 Analog modulation. The optical ports of the device under test are connected through short optical cables to an optical signal source and detector. The source shall produce a constant level, static signal easily detected by the detector. The output of the detector is monitored for analog modulation while the device under test is subjected to a test exposure. The monitoring equipment should include suitable transient capture capability (such as provided by an oscilloscope with an adjustable sweep trigger and oscilloscope camera).

4.6.2.5 Pulse distortion. The optical ports of the device under test are connected through short optical test cables to a suitable optical signal source and detector. The signal source shall be driven so as to provide a series of optical pulses having maximum rise and fall times of 1 nanosecond, and pulse durations of approximately 5 nanoseconds. The rise and fall times shall be measured between the 10 and 90 percent optical power levels, and the duration between the 50 percent levels. The transmitted optical pulses shall be compared with the incident optical pulses or other reference pulses known to be of the same shape as the incident pulses. The detector and monitoring equipment shall possess sufficient sensitivity and frequency response to suitably monitor both the incident, or reference, pulses and the transmitted pulses. The monitoring equipment shall also include suitable recording capability (such as provided by an oscilloscope and camera) so that a permanent record of the test data can be obtained.

4.6.2.6 Crosstalk. The optical ports of the device under test shall be connected to short test cables or opaquely capped as appropriate. The input port to one optical channel (active channel) is connected via its test cable to an optical signal source, the signal of which may be either continuous or amplitude modulated as appropriate. The output port(s) of the other channels (passive channel(s)) are connected to test cable(s). The other input ports shall be opaquely capped. The output signal power from both the active and passive channels shall be measured. When there is more than one passive channel, the output of all passive channels shall be summed. This test shall be repeated, making each remaining input port the active channel.

4.6.2.7 Ambient optical pickup. The optical ports of the device under test are either opaquely capped or connected to short optical test cables as appropriate. The far ends of these tests cables shall be either capped or connected to suitable optical power monitoring equipment. Mode stripping shall not be employed. The device under test shall be isotropically and homogenously bathed in simulated sunlight per method 505.1 of MIL-STD-810. The radiant energy source shall be broad spectrum with infrared, visible, and ultraviolet components and shall have an irradiance (power density) of 112 ±5 milliwatts per square centimeter. Optical power emanating from the optical ports of the device under test shall be measured by the optical power monitoring equipment.

4.6.3 Environmental test methods. Environmental test methods, as performed herein, shall meet all requirements of the applicable requirement(s) paragraph(s) as specified in table I for the particular test method.

4.6.3.1 Thermal shock. Penetrator assemblies shall be tested in accordance with condition A-1 of method 1003 of MIL-STD-1344 modified for a low temperature extreme of -40°C and a high temperature extreme of 71°C. Extreme temperature exposure durations in steps 1 and 3 of method 1003 as modified above shall be one hour. The unmated connector assembly shall be examined for satisfaction of the mechanical requirements of 3.7.1 and mated. Cleaning of the optical contacts of the unmated penetrator assembly in accordance with 4.6.3.20, and prior to optical testing shall be permissible.

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4.6.3.2 Temperature life. Mated penetrator assemblies and penetrator piece parts shall be tested in accordance with condition 3 and test time condition B of method 1005 of MIL-STD-1344. Assemble the piece parts and perform optical tests. The test assemblies shall meet the requirements of 3.7.2 following the test exposure.

4.6.3.3 Humidity. Penetrator assemblies (1 mated and 1 unmated) shall be tested in accordance with the type II, temperature cycling procedure of method 1002 of MIL-STD-1344. The unmated connector assembly shall be examined for satisfaction of the mechanical requirements of 3.7.3. Both test connectors shall be mated and shall be optically tested.

4.6.3.4 Salt spray (corrosion). The mated penetrator assemblies shall be tested in accordance with condition C of method 1001 of MIL-STD-1344. After test exposure the assemblies shall be externally cleaned, examined for salt penetration into the penetrator junction area, remated and optically tested to meet the requirement of 3.7.4.

4.6.3.5 Fluid immersion. Mated penetrator assemblies shall be tested in accordance with method 1016 of MIL-STD-1344 with the following modifications. Unless otherwise specified, each mated assembly shall be tested to a different immersion fluid called out in MIL-W-22759/43.

4.6.3.6 Flammability. The mated connector assemblies shall be tested in accordance with method 1012 of MIL-STD-1344. All requirements of 3.7.6 shall be met following the test exposure.

4.6.3.7 Ozone exposure. The mated connector assemblies shall be tested in accordance with method 1007 of MIL-STD-1344. After test exposure the subassemblies shall be examined for damage, mated and tested for optical performance as specified in 3.6. All requirements of 3.7.7 shall be met by these assemblies following the test exposure.

4.6.3.8 Hydrostatic pressure. The penetrator receptacles shall be tested on both sides in accordance with Test Condition A and Procedure 1 of Method 1006 of MIL-STD-1344. Pressure will be applied first to one side, decreased to zero, then applied to the other side. Following the pressure tests and after drying the exterior surfaces, the penetrator receptacle shall be examined for damage and shall meet all of the requirements of 3.7.8.

4.6.3.9 Mating durability. Mateable connector assemblies or accessories shall be tested in accordance with method 2016 of MIL-STD-1344. One hundred complete (part separating) mating and unmating cycles shall be accomplished at the maximum rate of one cycle per 15 seconds. The requirements of 3.7.9 shall be met.

4.6.3.10 Terminus retention. Unmated penetrator samples with or without removable termini, shall be tested in accordance with method 2007 of MIL-STD-1344. Termini shall be subjected to axial compressive loads applied at the front face of the optical terminal tending to push the terminus to the rear of the shell body. Axial loads shall be applied up to the maximum load specified. Terminus retention forces shall be measured after the first and tenth insertion or removal cycles in accordance with the maintenance aging requirements of method 2002 of MIL-STD-1344.

4.6.3.11 Cable seal flexing. Unmated penetrator plug subassemblies, of each type submitted shall be tested in accordance with method 2017 of MIL-STD-1344. After test exposure the subassemblies shall be examined for damage and shall meet all requirements of 3.7.11.

4.6.3.12 Twist. Mated penetrator assemblies shall be tested as follows. The penetrator assembly shall be clamped in a rotatable fixture capable of plus or minus 360 degrees of rotation. The assembly cables shall be stretched with minimum tension (about 5 kilograms equivalent force) to their maximum lengths and clamped at a distance of about 100 times the cable diameter from the connector to the test fixture table top. The penetrator holding fixture shall be rotated plus or minus 360 degrees at a rate of one cycle per 5 seconds for a total of 50 cycles. The test assemblies shall meet the requirements of 3.7.12.

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4.6.3.13 External bending moment. Mateable penetrator assemblies shall be tested to the following procedure. The cable receptacle shall be mounted as in normal service to a rigid panel. Before mating the cable plug to the receptacle, a bending moment test arm shall be secured to the rear of the plug shell. The fixture shall be of any convenient design for application of the load except it must not provide support for the connector shell in front of the engaged threads (see figure 1). A description of the fixture used shall be submitted to the contracting activity. After mating the plug and receptacle, the bending moment torque listed in table III and as measured from the panel shall be applied. The torquing load (P) shall be applied across the smallest exterior dimension (H) of the connector (see figure 1). The torque shall be applied at a rate of approximately 1 newton-meter per second until the required torque is applied. The maximum torque shall be held for 1 minute. The requirements of 3.7.13 shall be met following the test exposure.

TABLE III. External bending moment equivalencies.

Dimension H		Bending moment torque
More than (mm)	Equal to or less than (mm)	(newton-meter)
0	15	15
15	18	30
18	21	34
21	24	42
24	28	47
28	33	59
33	37	64
37	42	71
42	46	85

4.6.3.14 Cable pullout. Mated penetrator assemblies shall be tested in accordance with Test Condition C of method 2009 of MIL-STD-1344. The test assemblies shall meet the requirement of 3.7.14 following the test exposure.

4.6.3.15 Impact (drop). Unmated penetrator plug subassemblies of each type submitted shall be tested in accordance with method 2015 for light service, of MIL-STD-1344 with the following modifications:

a. The pipe and cable combined length (figure 1, method 2015) shall be two meters.

b. The exposed cable length (figure 1, method 2015) shall be 500 millimeters.

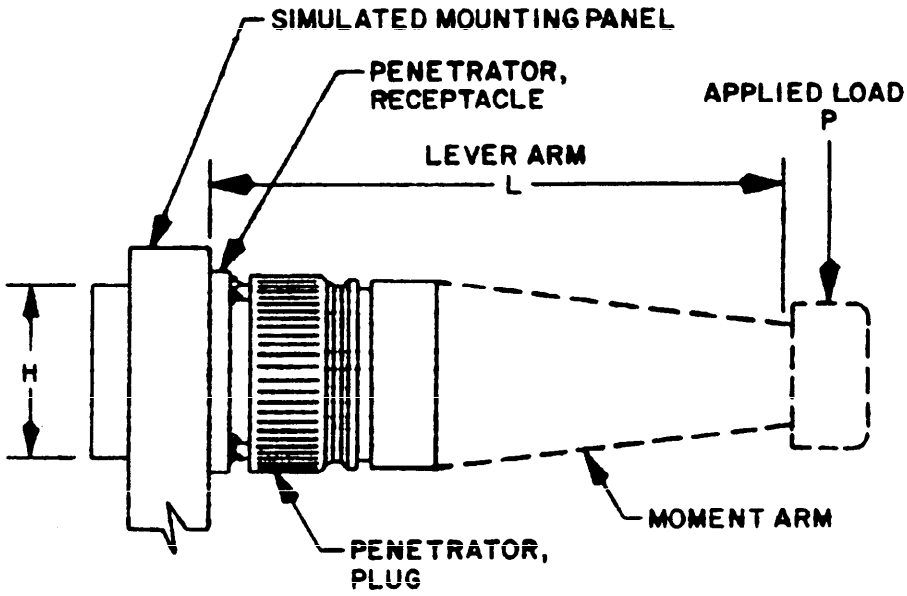
c. A total of 4 drops per assembly shall be made from a height of 1 meter above the floor. A rotation of 90 degrees of the penetrator assembly shall be made between drops.

The test assemblies shall be examined for damage, then mated and shall meet the requirements of 3.7.15 following the test exposure. Either cleanliness during testing will be required, or the contact cleaning procedure of 4.6.3.20 shall be employed after exposure.

4.6.3.16 Mechanical shock. Mated penetrator assemblies shall be tested in accordance with MIL-S-901, grade A, class I, type B. The test assemblies shall meet the requirements of 3.7.16 following the test exposure.

4.6.3.17 Vibration. Mated penetrator assemblies shall be tested in accordance with condition II of method 2005 of MIL-STD-1344. The test assemblies shall meet the requirements of 3.7.17.

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NOTES:
1. Moment arm fixture shall be on a convenient design.
2. Dimension H is the smallest exterior dimension of the connector.

FIGURE 1. External bending moment test setup.

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4.6.3.18 Terminus insertion and removal forces. Unmated penetrator subassemblies with removal termini, shall be tested in accordance with method 2012 of MIL-STD-1344. Ten insertion and removal cycles shall be performed in accordance with method 2002 of MIL-STD-1344, and the maintenance aging conditions specified therein shall be evaluated by measurement of the insertion and removal forces during the first, fifth and tenth aging cycles.

4.6.3.19 Water pressure. Penetrator assemblies shall be tested for water pressure susceptability by immersion in an aqueous dye penetrant solution to a depth of 2.0 meters for a period of 48 hours. The solution temperature shall be maintained between 10°C and 35°C during the exposure period. The dye concentration shall be adequate to visibly indicate liquid exposure. The penetrator shall be externally cleaned and examined for possible dye penetration into the seal region.

4.6.3.20 Terminus cleaning. The terminus cleaning procedure shall be supplied by the penetrator manufacturer. The visual examination requirement of 4.6.1 and insertion loss requirement as specified (see 3.1) shall be met following this terminus cleaning operation.

4.6.4 Physical property test methods.

4.6.4.1 Mass. Penetrators and accessories of each type submitted for testing shall be individually weighed and the measured masses shall be equal to or less than those specified (see 3.1).

4.6.4.2 Color. Penetrator color shall be compared visually for conformity with the color specified (see 3.1).

4.6.4.3 Identification marking. Visual examination shall be made of all marking characters on the tested penetrator assemblies or accessories. The requirement of 3.8.4 shall be met as a result of this examination.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-55330.

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1.1 Protective covers shall be provided for all connector surfaces.

6. NOTES

6.1 Intended use. The fiber optic bulkhead penetrators covered by this specification are intended for use in any navy ships and inboard submarine application where their performance characteristics are required within the limitations of their specified performance requirements. Water pressure is to be applied to the penetrator only one time.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title number, and date of this specification.

b. Type and size of penetrator required with reference to the applicable specification sheet number, title and date.

c. When a first article is required for inspection and approval (see 3.2 and 4.4).

d. Color required (see 3.8.3).

e. Total quantity of penetrators required.

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6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Insertion loss. The total optical signal power loss in a transmission system caused by the insertion of an optical component such as a bulkhead penetrator.

6.4.2 Discontinuity. Discontinuity is the presence of a relatively sharp, short duration interruption or "dropout" in the transmitted optical signal.

6.4.3 Analog modulation. Analog modulation is the dynamic, amplitude modulation of the optical signals transmitted by the optical device in question.

6.4.4 Pulse distortion. Pulse distortion is the change in waveform shape or duration of optical pulses transmitted by the optical device in question.

6.4.5 Crosstalk. Crosstalk is the coupled optical signal interference from one optical signal circuit into another presumed isolated optical signal circuit.

6.4.6 Ambient optical pickup. Ambient optical pickup refers to interference of the optical data signals in the device under test by ambient optical power incident upon its exterior.

6.4.7 Pressure-proof. Capable of withstanding (no mechanical or optical performance degradation) sea water at 7 MPa pressure.

6.5 Sub-contract material and parts. The preparation for delivery requirements specified in referenced documents listed in Section 2 do not apply when material and parts procured by the supplier for incorporation into the equipment lose their separate identity when the equipment is shipped.

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Review activities:
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User activities:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-24627(NAVY)	Penetrators, Bulkhead, Use on Navy Ships and Submarines) ~~Connectorized, Fiber optic.~~ (For Inboard

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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